3 **(a)** Determine, in terms of k, the coordinates of the point where the lines with the following equations intersect.

$$x+y=k$$

$$2x-y=1$$

[3]

(b) Determine, in terms of k, the coordinates of the points where the line $x+y=k$ crosses the curve $y=x^2+k$. **[4]**

3(a)	
	(answer space continued on next page)

3(a)	**(continued)**
3(b)	